Elixir cheat sheet

drum machine firmware for the M-VAVE FM-1 · 1 of 4

Black keys numbered 1–11 and white keys 1–16, left to right

A BEAT IN A MINUTE

HOME grid · black **1** (D1) · white 1 5 9 13 · black **4** (SN) · white 5 13 · black **6** (CH) · white 1 3 5 … 15 · **PLAY**. Turn **PRESETS** for kits, **OCT+** loads one.

KEYS AND KNOBS

White 1–16	Steps of the bar on screen · patterns (pattern page) · rolls and effects (PERF)
Black 1–7	Track: D1 D2 D3 SN CY CH OH (CH and OH share a voice)
Black 8–11	Bar 1–4 · banks on pattern, song, save and load pages · repeat length in PERF
OCT–	**SHIFT** (hold): the screen shows what SHIFT (OCT–) + each key does
OCT+	**COPY** (hold) · OK in dialogs · casts a spell
ALGORITHM	Select the track
SELECT	The page; in lists, the row
PRESETS	Kit browser; on a sound page, the sound browser
KNOB 1–4	The four cards at the bottom of the screen
MASTER	Output volume

BUTTONS (PRESS AGAIN FOR THE NEXT PAGE)

HOME	Grid → patterns → song
EDIT	Model → oscillator → FM → transient
ENV	Amp EG → gate → mod EG / velocity
SCL	Filter → shape → insert FX
LFO	LFO 1 → LFO 2 → random per hit → cross mod / choke
GLO	Mixer → config
FX	FX bus → reverb (SELECT for the pages)
SEQ	Track pages (track, generator, action, groove, bounce, melody, scale) → step edit
ARP	PERF: PERFORM → SPELLS → WANDER → GLOBAL MOD
SAVE	Save → load
PLAY · REC	Start / stop · record on / off

SHIFT (OCT–) + …

white 1–7	Play that track's voice
white 8 · 9	KIT: load a kit · MORPH: load the morph target
white 10 · 16	SONG page · PATTERN page
white 11 · 12	PROJECT: load a project · CONFIG
white 13	RELOAD the kit and pattern as saved
white 14	LAST: tap the track's last step (again: all tracks)
white 15	FX
track key	Mute / unmute
bar key	Track length 16 / 32 / 48 / 64
OCT+	**CLEAR**: SELECT a choice, SHIFT (OCT–) + OCT+ again to confirm
GLO	Solo / unsolo the selected track
HOME	PERF RESET: every performance effect off
EDIT	KEYS: the keys play the track's voice
SEQ	The links
PLAY	Play the song from its start
REC · SAVE	Record options · load page

COPY (OCT+) + …

bar key	Copy the shown bar there (on the grid: every track)
track, track	Copy a track
pattern, pattern	Copy a pattern (pattern page)
held + knob	Control All (sound page): the knob on every track; let go, then OCT+ reverts, OCT– keeps

LOCKS

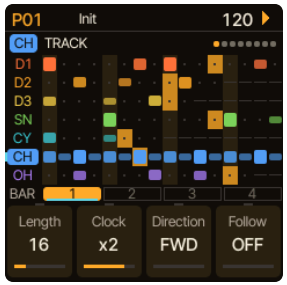
On a sound page, **hold a step key and turn a knob**: that value for that step only. Turning knobs while recording writes locks too.

THE GRID

Bar height: velocity · outline: chance under 100% · notch: flam · tick below: shift · dot top right: lock · dot top left: condition · bar below: ratchet. Knobs: tempo (down to MIDI: follow clock), shuffle, length, volume.

STEP EDIT (SEQ TWICE)

Tap a white key to select a step (again: on / off). SELECT for the pages.



Page	K1	K2	K3	K4
STEP	Note	Length	Velocity	–
TRIG	Chance	Condition	Shift	Flam
LOCKS	Lock 1	value	Lock 2	value
RATCHET	Hits	Shape	Rate	Fade

Length: AUTO, 1/32 … 1/2, 3/4 … 64 BARS, INF (a drone), TIE (slides into the next note). Shape: even or BOUNCE.

SOUND PAGES

Each page has two rows of four knobs: the button again (or SELECT) gives the other row. MODEL picks LXR, Rings or Plaits (D1–D3) or a MICRO type (any track).

SAVING (SAVE · SHIFT (OCT–) + SAVE LOADS)

KIT	64 slots or the library's 128 · OCT+, name, OCT+
PATTERN	32 · a white key saves to that pattern
SONG	64 · OCT+
MORPH	Saves the mixture you hear; loads as the morph target
SOUND	The library's 128: one track's sound
PROJECT	4 · everything
FACTORY	The 85 factory kits (load only)

Naming: K1 position, K2 letter, K3 case; OCT+ saves, OCT– alone goes back.

PATTERNS AND SONGS (HOME)

Pattern page: white key queues a pattern, bar keys pick the bank. Song page: K1 song, K2 row, K3 the row's pattern (END, LOOP), K4 repeats (x1 … x16).

MIDI

In: USB and the TRS jack · out: USB. Clock out always; tempo down to MIDI follows clock in. CC *n* sets parameter *n* – 1 (as on the LXR), NRPN reaches the rest. Program change 0–31 queues a pattern.

TRACK PAGES (SEQ ONCE, SELECT)

Page	K1	K2	K3	K4
TRACK	Length	Clock	Direction	Follow
GENERATOR	Generator	its settings		
ACTION	Action	Grids map X / Y, chaos · Euclid's second pulse		
GROOVE	Groove	Amount	Humanize	Follow
BOUNCE	Bounce	Bnc pitch	–	–
MELODY 1	Shape	Spread	Bias	Steps
MELODY 2	Deja vu	Loop	Length	Len var
SCALE	Root	Scale	Follow	Base

EACH TRACK ON ITS OWN CLOCK

Clock: /4 · /3 · /2 · 2/3 · 3/4 · **1x** · 4/3 · 3/2 · x2 · x3 · x4. A hat at x2 plays 32nds; 3/4 fits 12 steps in a bar. Add a length of its own (1–64) for polymeters.

FWD · REV	1 2 3 4 1 … · 4 3 2 1 4 …
PING	1 2 3 4 3 2 1 2 … (the ends once)
PING2	1 2 3 4 4 3 2 1 1 2 … (the ends twice)
RAND · WALK	Any step · one step either way

GENERATORS

EUCLID	Length, Hits, Rotate; a second pulse on ACTION (OR, XOR, AND, SUB)
MUTATE	Small changes each time round · Freeze stops it
TURING	Shift register: Length, Lock (127 repeats), Density
CELLULAR	Each time round the next row of rule 0–255
GRIDS	The Grids drum map: Part, Density, Accent

All but EUCLID play a live copy; COMMIT (ACTION page, OCT+) writes it into the pattern.

PERF (ARP): WHITE KEYS ACT WHILE HELD

SHIFT (OCT–) + key latches it. Black keys mute. Bar keys: repeat length 1/4 … 1/32. ALGORITHM picks the throw track.

Key	PERFORM	PERFORM 2
1–7	Roll the track	DRILL: random fast ratchets
8	FILL	HLF: half time
9	RPT: repeat	JIT: jitter
10	FRZ: freeze the delay	CUT: random cuts
11	T.RV: throw to reverb	SHT: decays short
12	T.FX: throw to FX bus	PMP: pump off D1
13	FB+: feedback push	RES: comb resonator
14	STP: tape stop	SPC: reverb pad
15	LP~: low-pass sweep	CRS: crush
16	HP~: high-pass sweep	REV: reverse

PERF PAGES (SELECT)

Page	K1	K2	K3	K4
PERFORM	Tempo	Roll rate	Morph	SRate
MORE	DJ filter	Repeat	Shuffle	Tape time
DENSITY	Density	Ghost vel	Build	Vary
STUTTER	Chance	Size	Length	Vary
RUSH	Rush	Top	Filter	Glide
RUSH SETUP	Roll at	Free at	Flam	Pitch
TRANSPOSE	Degrees	When	Keys	–
2: SETUP 1	Drill	Jitter	Cuts	Pump
2: SETUP 2	Comb	Comb fb	Crush	Reverse

Hold a roll key and turn K2 on PERFORM: a sweep from 1/4 to 1/192.

GLOBAL MOD (ARP FOUR TIMES)

LFO 1–3	Dest, Depth, Speed (1/16 … 256 bars), Wave · Mode, Trig, Phase
MACROS	Four knobs, each a parameter between Min and Max (MACRO SETUP)
LINKS	Eight couplings between tracks: When, Target, Effect, Window

DENSITY (PERFORM: DENSITY)

Density	–100% thins the groove, weakest steps first (off-beat 16ths, then 8ths, quiet before loud); +100% raises chances, then adds
Ghost vel	Velocity of the added notes
Build	What + adds: CHANCE, GHOSTS, RATCHETS
Vary	OFF the same steps go first; up, a new order each time round

Black keys leave a track out. OCT+ commits it into the pattern.

STUTTER (PERFORM: STUTTER)

Chance	Chance of a burst at each Length (0%: only OCT+ and LFOs)
Size	The slice: 1/8 … 1/64, triplets, RANDOM
Length	How long a burst lasts: 1/16 … 1 bar
Vary	OFF, SHRINK, PITCH DN / UP, REVERSE, RANDOM

OCT+ held on the page stutters now. It repeats the sound itself, tails and reverb included.

LINKS (SHIFT (OCT–) + SEQ)

Up to 8 per pattern, read as a sentence: *When D1 hits, CH chance –60%, on the next step.*

When	A track that hits, accents (vel. over 100) or rests
Effect	DUCK, ONLY, TRIG, RATCH, VEL, PITCH, SHIFT, DECAY
Window	SAME step, NEXT, NEXT 2, the rest of the BAR

MORPH

SHIFT (OCT–) + white 9 loads a second kit as the target; PERFORM's K3 crossfades every sound parameter to it (switches flip halfway). M in the header while it's above 0.

MIXER AND FX

GLO	Volume, pan, FX send, reverb send (2 pages each)
FX send	How much of the voice goes through the FX bus
Reverb send	Feeds the reverb on top of the output
FX	FX bus (delay with TAPE modes, dirt, compressor) → reverb

SPELLS (ARP TWICE)

Changes the sound and pattern of the tracks you pick, at random but in a direction.

1 Targets	Black keys (bar key 1: all / none)
2 Parts	White keys: what may change (below)
3 Spell	K1 on GRIMOIRE
4 Cast	COPY (OCT+); again: a new roll from before the first
5 Undo	SHIFT (OCT–) alone: the pattern and kit as before
Bar keys	1 all / none · 2 keep · 3 save a recipe · 4 knobs reset

THE KNOBS, PAGE BY PAGE

GRIMOIRE: which spell

Spell	NONE, the eleven spells, then your recipes U1–U8
Variation	EXACT casts the knobs as set; higher nudges them a little each cast
Scope	White-key presets: ALL, SOUND, PATTERN, VOICE, SHAPE, SPACE …
Roll	NEW each time; SAME repeats the last roll elsewhere

CAST: how much

Sound	How far the voices move; 100% (FREE) frees the kick and toms
Pattern	How much the steps change (OFF: none)
Density	Fewer hits (left) or more (right); AS IS in the middle
Chaos	Low follows the beats; high anything goes

CHARACTER: which way (names: next column)

K1	The sound's character
K2	The rhythm's character
K3	Evolution: how much keeps changing after the cast
K4	Protection: keeps kick, snare, key, main pulse

LIMITS: the most any spell sets (shared, saved)

Sends	Rev Send, FX Send: cap each target's sends
Levels	Rev Level, FX Mix: cap the shared FX

MOTION: how far the tracks part (needs 15 LEN)

Motion	SAME · ONE track at its own speed or direction · FEW · MANY · FREE
Lengths	BARS 16/32/64 · EVEN +12, 24 · ODD +7, 14, 15, 20

THE SPELLS AND THEIR CHARACTER KNOBS

None	Plain chance: Sound from a nudge to anything	Spread, Detail, Motion, Protect
Poltergeist	Metallic, unstable hits and rattles	Metal, Rattle, Haunting, Preserve
Skeleton	The groove stripped to its bones (pattern only)	Emphasis, Syncope, Variety, Anchors
Glitch Storm	Digital errors, bursts, cuts, fills	Corrupt, Fragment, Unstable, Groove
Haunted Dub	Dark filters, echoes, space; kick dry	Darkness, Echo, Drift, Dry Kick
Forbidden Ritual	All at once: models, steps, links, FX	Alchemy, Disorder, Possess, Protect
Raster Séance	Microsound clicks, grains, silences	Grain, Fragment, Jitter, Backbone
Concrete Drift	Minimal changes over many bars	Texture, Motion, Drift, Stability
Rust Circuit	Industrial metal, drive, cycles	Corrode, Machine, Malfunct, Pulse
Glass Cathedral	Resonant, tuned, glassy, in one key	Brilliance, Harmony, Shimmer, Tuning
Biofeedback	Tracks answering each other	Coupling, Response, Adapt, Freedom
Whispering Motes	Soft microsound dust and chirps	Grain, Scatter, Whisper, Anchors

The screen says what a CHARACTER knob does as you turn it. D1 stays a kick and D2, D3 toms unless Sound is 100%. SHIFT (OCT–) + a white key: only that part.

PARTS (WHITE KEYS)

Sound (white 1–8)		Pattern (white 9–16)	
1 PIT	pitch	9 HIT	which steps play
2 OSC	osc, FM, models	10 ACC	accents
3 TRN	transient	11 PRB	chance, conditions
4 ENV	envelopes, level	12 TIM	timing, flams
5 FLT	filter	13 RAT	ratchets
6 DRV	drive, shaper, bits	14 NOT	notes
7 FX	sends, pan, insert	15 LEN	length, clock, direction
8 MOD	random, LFO 2, xmod	16 LNK	links, generators

WORTH KNOWING

Lost? SHIFT (OCT–) + HOME (PERF RESET) turns every held, latched or running performance effect off. The LFOs, morph and macros stay.

Control All lets you back out. Hold OCT+ on a sound page and turn: every track moves. Let go and it asks: OCT+ reverts the whole kit, OCT– keeps it.

Generators don't touch your pattern (but EUCLID does): MUTATE, TURING, CELLULAR and GRIDS play a live copy. COMMIT keeps what you hear; RESET starts over.

Clocks change time, not steps. A track at 3/4 still has 16 steps; they just take 4/3 of a bar. Give it 12 steps to fit a bar.

KEYS stays on as you change pages: play the voice while you turn knobs. The header shows KEYS. SHIFT (OCT–) + a key plays at the accent velocity.

Recording: REC, then PLAY. Knob moves on sound pages become locks on the steps that are on; SHIFT (OCT–) + CLEAR erases the track's steps as they pass.

Morph and edits: with morph up, a knob moves the target too, so you hear the edit at any amount (CONFIG → MORPH to change it).

Solo isn't saved and the mutes come back as they were when the last solo goes.

Kits: PRESETS lists the 85 factory kits, then the library, then the project's 64 slots. The library stays whatever project is loaded.

VOICES (ON MOTION'S PAGE, WHITE 1–5)

The engines a cast may use: **1 LXR** · **2 RNG** Rings · **3 PLT** Plaits · **4 MIC** MICRO · **5 KCK** D1 the LXR's kick. All lit: the spell picks; none: engines kept.

WANDER (ARP THREE TIMES)

The pattern changes slowly, over minutes, until you are somewhere else. Patterns are never changed for good: the one you leave gets its steps back.

1 Tracks	Black keys (bar key 1: all / none)
2 Parts	White keys: what may change (SHIFT (OCT–) + white: only that one)
3 Quest	K1 on QUEST, how long with Journey
4 Set out	COPY (OCT+): here is home (the pattern, the voices, the morph kit)
5 Home	SHIFT (OCT–): back over Return bars; again: at once

Page	K1	K2	K3	K4
QUEST	Quest	Journey	Fate	Goal
PACE	Sound	Rhythm	Range	Stride
HOME	Return	Memory	Order	Sway

Sound	Bars a voice moves a notch (OFF: rhythm only)
Rhythm	Bars an Energy / rhythm step, at the pattern's start
Range	How far Energy swings (thin to twice as busy)
Stride	Step size, threads at once (1–3)
Goal	NONE (open road: the steps change), P01–P32 (walk there), SONG (the song's rows)
Fate	HOME · TREASURE (one change kept) · WOUND · LOST (here is home) · DICE
Memory	How often the road returns to a version you heard
Order, Sway	Goal SONG: the next row (IN ORDER, ROADS, ANY); how much the time varies

WANDER'S BAR KEYS (BLACK 8–11)

black 8	Every track, or none
black 9	**KEEP**: here is home now
black 10	**REST**: the road waits; again, on
SHIFT + black 10	**POSTCARD**: where you are into the first empty pattern, its voices LOCKED (SAVE to keep it)
black 11	Wander's knobs and keys as at first

THE QUESTS

ROAD one steady walk · **ROUND** out and back · **SPIRE** climb, summit, down · **HOLLOW** into the cave, to a pulse, out · **TIDE** ebb and flood · **DETOUR** side quests · **ASTRAY** anywhere, never home · **MAZE** by rule · **ROSARY** hits leave one by one, then return · **POND** the slowest · **WHEEL** an endless, rhythm-only beat

Rhythm under a morph: Sound OFF (or WHEEL), OSC/MODELS out, and a global LFO on MORPH sweeps between two kits while the road changes the steps.

A spell cast on the road is a side journey: reverse it and the road goes on. Coming home, each melody starts again from its top.

RUSH (PERFORM: RUSH)

Black key: the rush track. Turn **Rush**: the other drums go over to it (hats first, kick last), it rolls on the grid (16ths, 32nds), then free up to Top (1/64–1/256); the high-pass opens at the top. Slow turns 1% a detent, fast spins 10%.

REC + turn: RUSH locks in the steps. On RUSH pages the white keys are the steps: hold one and turn to lock it.

TRANSPOSE (PERFORM: TRANSPOSE)

White keys: –7 … +8 degrees of the scale (the 8th, breathing, is 0). Black keys: which tracks move. When: NOW, BEAT, BAR. Keys: LATCH or HOLD. Not saved.

BASE AND NOTE LOCKS

Base (SCALE, K4): the track's base note, ±2 octaves; every note follows it, but a step whose **Note** you turned on STEP (it reads Locked).

LONG NOTES

STEP's **Length**: AUTO, 1/32 … 1/2, 3/4, 1 BAR … 64 BARS, **INF** (a drone till the track's next note), TIE. Holds a voice with Gate ON (amp EG's next page).

ARPS (MELODY: SHAPE)

ARP UP, DN, U/D, then CONVERGE 1-4-2-3 · DIVERGE 2-3-1-4 · CON/DIV · LEAPFROG 1-3-2-4 · SPIRAL 1-2-4-3 · ALTERNATE · PAIRS 2-1-4-3 · ROTATE 1234 2341 · INVERT 1234 4321 · PEDAL 1-2-1-3-1-4. Over the chord tones, Spread wide.

ROADS (SONG, THIRD PAGE)

K1	K2	K3	K4
Song	Way: PLAY / ROADS	Surprise	Home

ROADS plays the song by its own chances: after a row, the row after another with the same pattern. Surprise lets in the rarer turns; Home brings it back to row 1 every so many bars.